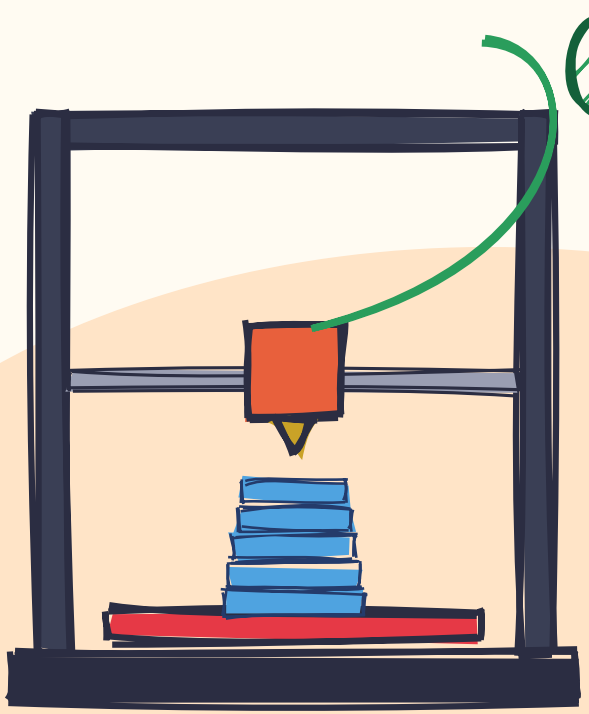
Foundation · For every age group

3D Modelling & 3D Printing

From an idea in your head to an object in your hand

Everyone · Ages 8+ · Ages 12+ · Ages 15+



How to use this deck

Coloured badges show the level of each slide. Teach 'Everyone' slides to all, then add the levels that fit your class.

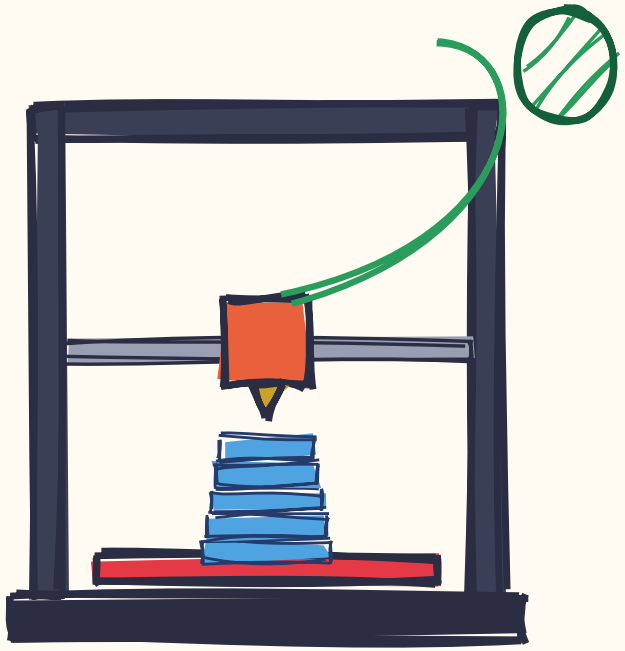
LEARN

Let's wonder...

What if you could MAKE anything?

A design superpower

A broken toy part, a phone stand, a robot arm, a name badge. Today, engineers draw these on a computer and a machine builds them **layer by layer** while they watch.



Buy it?

Carve it from wood?

Design and print it!

Today: you learn how 3D printing works, how to design your own objects, and how to print them **safely and well**.

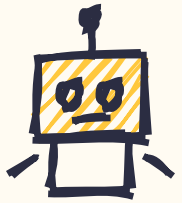
Learning aims

Today I will learn to...

1. Explain how a 3D printer builds objects **layer by layer**
2. Design a model in **Tinkercad** with shapes and holes
3. Measure in **millimetres** and design parts that **fit**
4. **Slice** a model and choose good print settings
5. Print **safely**, fix common problems and design responsibly

Keywords

3D model · CAD · STL · slicer · layer · infill · support · tolerance



LEARN

Watch and think

Video time!



Tinkercad Basics for Kids: Build 3D Models

Wize Computing Academy · about 5 minutes · youtu.be/8ccldtpWMmY

Watch for...

1. How do you put a **shape** on the workplane?
2. How do you change its **size**?
3. What does a **hole** shape do?

Scan to watch

After the video: Show your partner, with your hands, how you would make a cup from a solid and a hole.

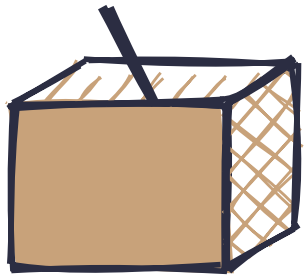
LEARN

Everyone

Big idea

Adding, not cutting away

Subtractive (old way)



carve away

Start with a block, cut away what you don't need. Lots of **waste**.

Additive (3D printing)



add layers

Build up thin layers, only where needed. Very **little waste**.

Like stacking pancakes

Each layer is about **0.2 mm** thick, thinner than 2 sheets of paper. A 2 cm tall object has about **100 layers**!

Try it

Stack coins of shrinking size to build a mountain. That is how a printer thinks!

Real world

What do people 3D print?



Prosthetic hands
Volunteers print low-cost hands for children around the world



Rockets & planes
Engine parts too complex to make any other way



Medicine
Dental braces, hearing aids and models of organs for surgeons



Prototypes
Engineers test a design in hours instead of weeks

? If you had a 3D printer at home, what is the FIRST useful thing you would print?

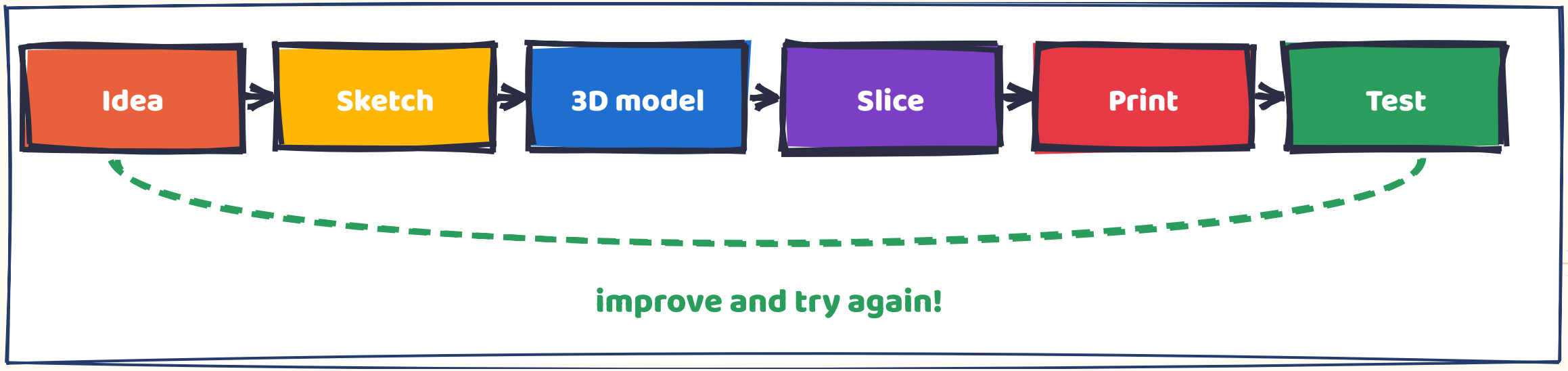
Even houses!
Giant printers can print the walls of a house from concrete in a few days.

LEARN

Everyone

The journey

From idea to object in 6 steps



Step	Tool	Typical time
Sketch	paper and pencil	2 minutes
3D model	Tinkercad (CAD software)	20 to 40 minutes
Slice	UltiMaker Cura makes printer instructions (G-code)	2 minutes
Print	the 3D printer	30 minutes to many hours

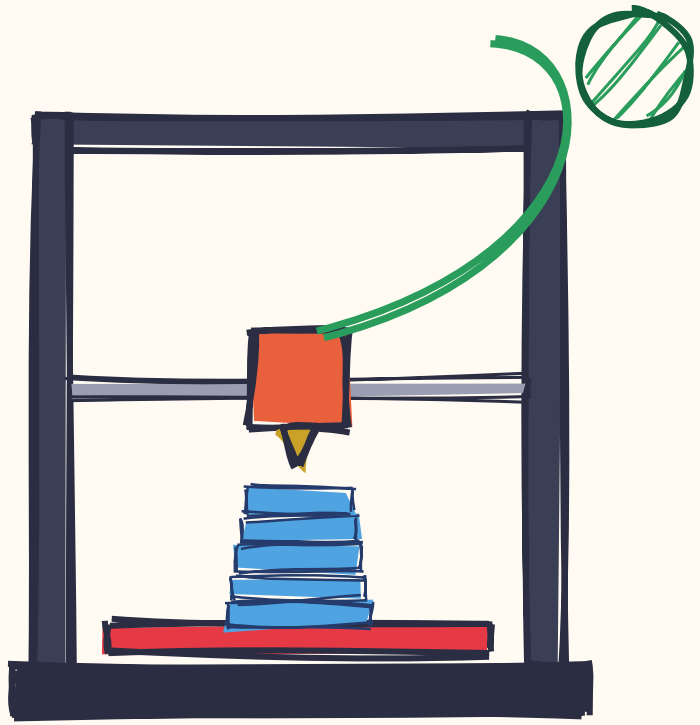
Patience!
Printing is slow. Plan so prints run overnight or between classes.

LEARN

Look inside

Everyone

How an FDM printer works



Part	Job
Filament	plastic string on a spool (1.75 mm thick)
Extruder	motor that pushes filament into the hot end
Nozzle	melts plastic at about 200 °C, hole 0.4 mm wide
Bed	flat plate the print sticks to, often heated to 60 °C
Motors	move in X, Y and Z very precisely

FDM
Fused Deposition Modelling: melted plastic laid down (deposited) in lines.

Like a hot glue gun on a robot

The nozzle draws one layer, the bed moves down a tiny bit, and it draws the next layer on top.

Materials

Which plastic should we use?

Filament	What it's like	Nozzle temp	Use for
PLA	easy, low smell, made from plants like corn	about 200 °C	our choice
PETG	tougher, a bit bendy, water-resistant	about 235 °C	outdoor parts
TPU	rubbery and flexible	about 220 °C	phone cases, tyres
ABS	strong but smelly, warps easily	about 240 °C	experts, with ventilation

Cost check

1 kg roll of PLA ≈ **Rs 4,600**
= about **Rs 4.6 per gram**.

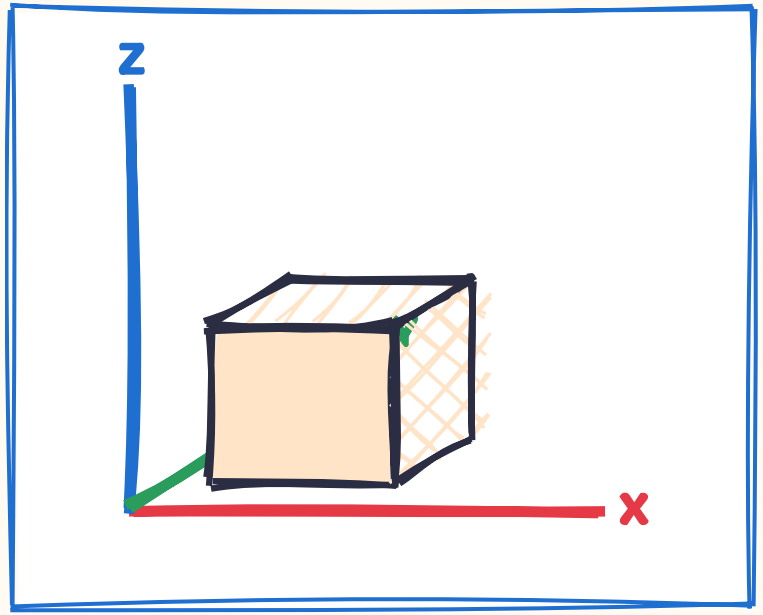
Keep it dry
Store filament in a bag with silica gel. Damp filament prints badly.

LEARN

Everyone · Ages 8+

Thinking in 3D

X, Y and Z, measured in millimetres



X = width
left and right

Y = depth
front and back

Z = height
up and down

Millimetres (mm)

3D design always uses **mm**. 10 mm = 1 cm. A coin is about 2 mm thick; a bank card is about 54 mm × 86 mm.

Measure tools
Ruler for big parts. A digital caliper measures to 0.1 mm!

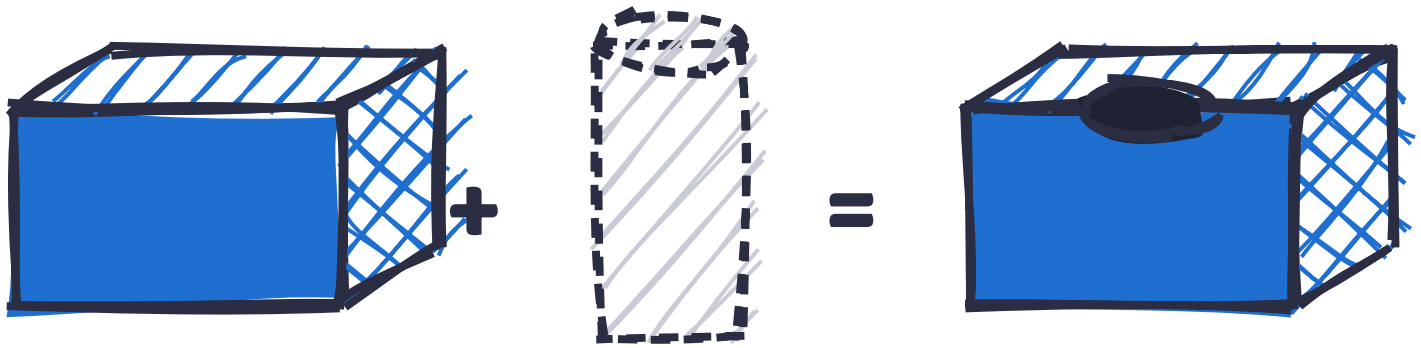
Younger groups: just use “wide, deep, tall”. **Older groups:** practise reading X, Y, Z coordinates in Tinkercad.

LEARN

Everyone

The magic trick

Solids + holes = any shape



Think in pieces

A cup = cylinder minus a smaller cylinder. A keyring = flat shape minus a small hole.

How to do it

1. Place a shape. 2. Place another shape and set it to **Hole**. 3. Select both and click **Group** (Ctrl + G). The hole is cut out!

Ungroup

Made a mistake? Select and Ungroup (Ctrl + Shift + G).

LEARN

Ages 8+ Ages 12+

Precision

Align, ruler and exact sizes

Align tool (L key)

Select shapes, press **L**, click a dot to line up their centres or edges perfectly.

Ruler (R key)

Drop the ruler on the workplane to type **exact distances** and positions in mm.

Type sizes

Click a white handle and **type** the number, e.g. 40 mm, instead of dragging.

Snap grid

Bottom-right setting. Set it to **1 mm** (or 0.5 mm) for neat designs.

Duplicate & repeat

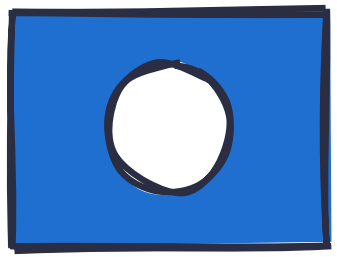
Ctrl + D duplicates AND repeats the last move. Great for rows of holes!

LEARN

Ages 12+ Ages 15+

Make parts fit

Tolerance: leave a little gap



hole 10.4 mm



peg 10 mm

The problem

A 10 mm peg will **NOT** fit a 10 mm hole. Melted plastic spreads a little, so holes print **smaller** and pegs print **bigger**.

Fit	Make hole bigger by
Loose / sliding	+0.4 to 0.5 mm
Snug push fit	+0.2 to 0.3 mm
Tight press fit	+0.1 mm

Test coupon

Before printing a big part, print a small strip with holes of 10.0, 10.2, 10.4 and 10.6 mm. See which fits best on **your** printer.

Measure twice

Use a caliper on the REAL part (sensor, motor, screw) before designing.

LEARN

Maths moment

Ages 8+ Ages 12+

How long, and how much?

Object	Filament	Print time	Cost (Rs 4.6 / g)
Name badge	8 g	35 min	8 × 4.6 = **Rs 37**
Sensor mount	15 g	1 h 10 min	15 × 4.6 = **Rs 69**
Plant monitor case	45 g	3 h 30 min	45 × 4.6 = **?**
Class of 30 badges	30 × 8 g	?	**?**

Plan the queue

One printer, 30 students, 35 minutes each = **17.5 hours** of printing! Print several small items **together** on one bed.

Answers

Rs 207 · 240 g, about Rs 1,104.

Costs use our lab PLA price (about Rs 4,600 per 1 kg roll). Times and grams are examples: always read the real numbers from Cura.

LEARN

Ages 12+ Ages 15+

Fix it

Common print problems

What you see	Try this
Print won't stick to the bed	level the bed, clean it, add a brim, check first layer height
'Spaghetti' mess	print came loose: same fixes, and slow down the first layer
Corners lift up (warping)	brim, heated bed on, avoid drafts and fans
Thin hairs between parts (stringing)	lower the nozzle temperature a little, dry the filament
Gaps or weak layers	nozzle partly blocked or filament tangled: ask the teacher
Overhang droops	add supports or rotate the model

First layer
80 % of problems start in the first layer. Watch it go down!

Make responsibly

Think before you print

Waste less: check the design and the Cura preview before printing. Every failed print is plastic waste.

Credit others: remixing a design from a sharing site? Follow its licence and name the designer.

Never print: weapons or parts that copy real ones, or anything meant to harm.

Respect brands: don't copy logos or famous characters for selling.

Ask: every print in our lab is approved by the teacher first.

Recycle

Keep failed PLA prints in a box. Some companies can recycle PLA into new filament.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Printers are hot and full of moving parts
2. Tinkercad: free 3D design in the browser
3. Rules that save failed prints
4. Export STL, then slice in Cura
5. The 5 settings that matter most
6. Design sprint!
7. Level-up challenges

MAKE

Safety first

Printers are hot and full of moving parts

Hot nozzle: about 200 °C. Never touch it, even after printing: it stays hot for minutes.

Hot bed: about 60 °C. Wait until it cools before removing prints.

Moving parts: keep fingers, hair, sleeves and ID cards away while it runs.

Fresh air: print in a ventilated room. PLA is mild, other plastics can smell.

Adults operate it: students design; a trained adult loads filament, starts and removes prints.



Never leave it alone

Check on long prints regularly. Keep the area clear of paper and clutter.

MAKE

Everyone

Our tool

Tinkercad: free 3D design in the browser



Drag shapes
box, cylinder, sphere, text



Move
drag, or arrow keys



Rotate
curved arrows on the shape



Resize
white corner handles; type exact mm



Look around
view cube top-left, right-drag to orbit



Undo
Ctrl + Z is your best friend!

Accounts for students

The teacher creates a **Tinkercad Classroom**. Students join with a **class code and a nickname**, so under-13s do not need their own email.

Website
tinkercad.com · works on school laptops, no install needed.

MAKE

Ages 8+ Ages 12+

Design for printing

Rules that save failed prints

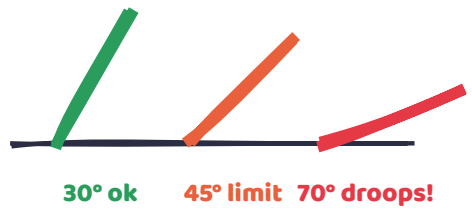
Flat base: the bottom must sit flat on the bed.

Overhangs: slopes steeper than 45° need supports underneath.

Thick walls: at least 1.2 mm (3 nozzle widths), or they break.

Text: raise or cut letters at least 1 mm and make them big.

Tiny parts: anything under 2 mm is fragile. Keep it chunky!



Rotate it!

Often the best fix is to lay the model on a different side.

MAKE

Ages 8+ Ages 12+

Prepare to print

Export STL, then slice in Cura

1. Export

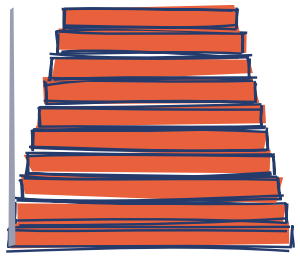
In Tinkercad: **Export → .STL**. An STL describes the surface of your model as thousands of tiny triangles.

2. Slice

Open the STL in **UltiMaker Cura**. Pick the printer and PLA, then click **Slice**.

3. G-code

Cura writes **G-code**: thousands of move and heat commands. Save it to the SD card or USB.



layer 10 of 10

Preview!

Use Cura's **Preview** slider to watch every layer before you print. Spot problems early, for free.

Cura shows print time and grams of filament. Use them on the cost slide!

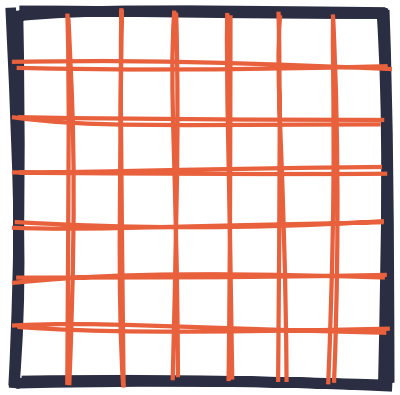
MAKE

Ages 12+ Ages 15+

Slicer settings

The 5 settings that matter most

Setting	Start with	What it does
Layer height	0.2 mm	smaller = smoother but slower
Infill	15 to 20 %	the inside pattern; more = stronger, slower, heavier
Walls	3 lines (1.2 mm)	outer shell strength
Supports	only if needed	props under overhangs; snap off after
Adhesion	skirt or brim	brim = wide edge that stops corners lifting



grid infill (top view)

100 % infill?
Almost never! It wastes plastic and time for little extra strength.

Class game

Design sprint!

How to play

1. Teams get a real problem card: e.g. **'My pencils keep rolling off the desk.'**
2. 5 minutes: sketch 3 different ideas on paper.
3. Choose one and model it in Tinkercad (20 minutes).
4. Present: what problem, what design, why it will work.
5. Class votes on which designs to print first.

Then discuss

Which design solved the problem in the simplest way? Simple usually wins!



Big idea

Good design starts with a real problem and a real user, not with the printer.

Level up!

Level-up challenges

Choose your challenge

Everyone: design a keyring with your initials that is under 10 g of plastic.

Ages 8+: design a box with a lid that fits (use a 0.4 mm gap).

Ages 12+: design two parts that click together with a snap-fit hook.

Ages 15+: use Tinkercad Codeblocks or OpenSCAD to generate a design from code.

Think like an engineer

Why do engineers usually print a cheap, fast version first instead of the perfect final part?

Boss level

Print a 'draft' at 0.3 mm layers and 10 % infill to test the fit in half the time.

Projects for every age

What each group will design

Little Explorers
ages 5 to 7

Teacher-led: stack shapes into a **robot** or **name keyring**. Teacher exports and prints. Focus: shapes and colours.

Junior Makers
ages 8 to 11

Name badge, then a measured **sensor mount** for the robot (JM-10). Focus: holes, align, mm.

Young Innovators
ages 12 to 14

Plant monitor enclosure with cable holes (YI-16). Focus: tolerances, slicing, cost.

Tech Leaders
ages 15 to 17

Parametric robot gripper in Onshape or Fusion (TL-15). Focus: parameters, strength, iteration.

Same rule for everyone

Design → print → **test on the real thing** → improve. Version 2 is always better than version 1!

Showcase

Display version 1 next to the final version. Visitors love seeing the improvement!

Show what you know

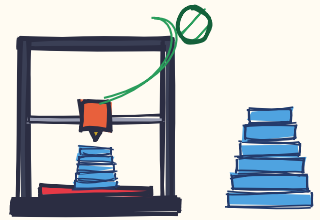
What did we learn?

- [x] 3D printers add plastic layer by layer (additive)
- [x] Tinkercad: solids + holes, grouped, measured in mm
- [x] Design rules: flat base, 45° overhangs, thick walls
- [x] Leave a tolerance gap so parts fit
- [x] Slice in Cura, print safely, test, improve

Exit ticket: why won't a 10 mm peg fit a 10 mm printed hole?

At home

Find 3 plastic objects at home. Could any be 3D printed? Which side would you print it on?



Next class: your group's 3D design class!